

Sazzad Hossen

Dhaka, Bangladesh

B.Sc. in Electrical & Electronic Engineering

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LinkedIn Profile

EDUCATION

B.Sc. in Electrical & Electronic Engineering (Major: Electronics, Minor: Communication) Feb 2022 – Feb 2026

United International University (UIIU)

CGPA: 3.92 / 4.00

Higher Secondary Certificate (HSC)

2020

Government Science College

GPA: 5.00 / 5.00

Secondary School Certificate (SSC)

2018

Shiruail High School

GPA: 5.00 / 5.00

WORK EXPERIENCE

Teaching Assistant

Mar 2026 – Present

United International University (Dept. of EEE)

Dhaka, Bangladesh

- Conducted lab demonstrations, led tutorials, and provided student counseling.
- Assisted faculty with course management and exam invigilation.
- Supported BAETE accreditation and coordinated departmental seminars.

Undergraduate Assistant

Mar 2025 – Jul 2025

United International University (Dept. of EEE)

Dhaka, Bangladesh

- Assisted faculty during lab classes and conducted extra tutorial sessions for difficult coursework.
- Provided student academic counseling and supported BAETE program accreditation activities.

Undergraduate Grader

2022 – 2024

United International University (Dept. of EEE)

Dhaka, Bangladesh

- Evaluated assignments and lab reports for multiple EEE courses.
- Maintained grading records and performance analysis using Excel.

CAPSTONE/THESIS PROJECT

- **Design of a 2.45 GHz RF Receiver in CMOS 45nm technology** : Designed and simulated a complete RF receiver chain by seamlessly integrating BPF, LNA, Mixer, and LPF blocks to optimize signal processing and minimize system noise. Achieved an overall conversion gain of 50 dB with a down-converted intermediate frequency (IF) of 10–20 MHz

ACADEMIC PROJECTS

- **Smart Glove for Sign Language Translation**: Sensor-based embedded system translating hand gestures into real-time text and speech.
- **Design & Layout for a 3,500 Sq. Ft. Apartment in a 10-Story Building**: Completed full electrical wiring, load calculation, and power distribution layout using AutoCAD.
- **License Plate Recognition (MATLAB)**: Image processing algorithm for automatic vehicle plate detection.
- **Design of Schematic and Layout of 8-bit Carry Save Adder** : Completed schematic simulation, functional verification, and layout implementation ensuring DRC & LVS clean.
- **Design of 2-Stage OTA in 45nm CMOS Technology**: Designed and simulated a high-gain, wide-bandwidth OTA using CMOS 45nm technology.
- **32-bit ARM ALU (Verilog)**: Developed RTL code for a 32-bit ALU featuring arithmetic, logic, and shift operations with full NZCV flag logic.
- **DSB-SC Signal Transmission & SDR Power Measurement**: Measured indoor path loss using a 915 MHz DSB-SC signal and SDR-based power analysis.
- **2 kW Power System Design (ETAP)**: Conducted load flow analysis and designed a 2 kW power system for an IEEE-14 bus network.
- **EcoBridge Smart Waste Management**: Developed a smart waste management and resource recovery solution for sustainable environments.
- **BPSK Modulation & Demodulation**: Simulated and successfully implemented BPSK signal modulation and demodulation in hardware.

SKILLS

Programming: C, Verilog, MATLAB

EDA Tools: Cadence Virtuoso, NCSim, ETAP

Simulation & Design: Proteus, Multisim, AutoCAD

Hardware: STM32, Arduino Uno

Soft Skills: Communication & Presentation, Leadership, Teaching & Mentoring

ACHIEVEMENTS

Champion VLSI Design Competition (EEE Day 2025)

2025

Champion Junior Intra-University VLSI Design Competition

Champion MATLAB: Simulate & Innovate

Runner-Up VLSI Design Competition, MIST Robolution

Runner-Up EEE Project Show (Senior Category)

Scholarship University Award for Academic Excellence

PROFESSIONAL DEVELOPMENT

Workshops & Training: Professional Skills Development, Workshop on Basic VLSI Design, MATLAB: Simulate & Innovate, Foundations of Real-World Mathematics, Public Speaking & Presentation Skills, Introduction to Computer Networks, DevOps for Engineers.

MEMBERSHIPS

UIU Electrical & Electronics Club, Govt. Science College Science Club, Shiruail High School Science Club.

Dr. Raqibul Mostofa

Professor & Dean, School of Science & Eng.

Director, Biomedical, Image & Signal Research

United International University, Dhaka

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Relation: Undergraduate Capstone Project Co-Supervisor

Dr. Md. Hasanuzzaman

Associate Professor, Dept. of EEE

United International University, Dhaka

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Relation: Undergraduate Capstone Project Co-Supervisor